Optimizing Retail Efficiency: A Data-Driven Approach for LAXMI GENERAL STORE

A Midterm Report for the BDM capstone Project

Submitted by

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Executive Summary :

This project focuses on Laxmi General Store, a well-established grocery store located in the vibrant area of Preetam Nagar, Prayagraj, serving the community within the B2C sector. The store is known for offering a wide range of products, including groceries, fresh produce, beverages, toiletries, and household essentials. Positioned as a convenient solution for daily household needs, Laxmi General Store is dedicated to providing high-quality products and enhancing the shopping experience for its customers.

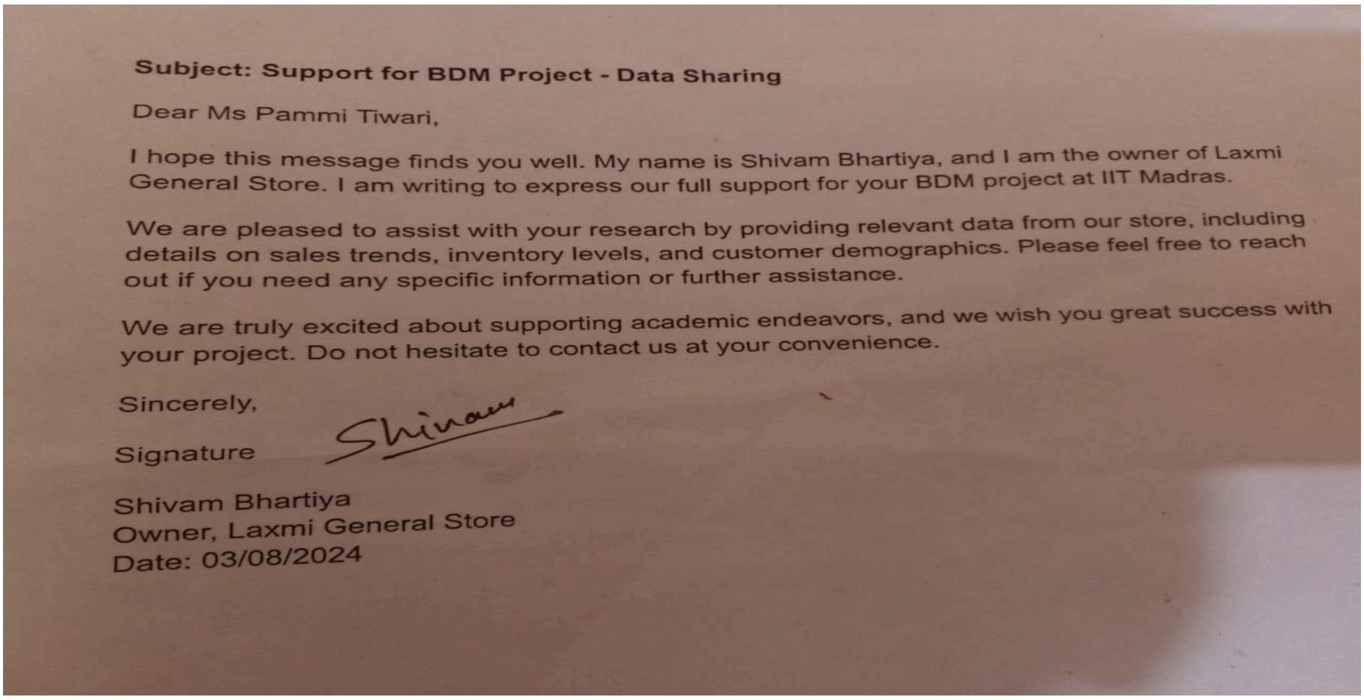
However, an in-depth analysis of the sales data has revealed two significant challenges impacting the store's profitability: excessive inventory and inefficiencies in inventory management. The large stock levels tie up crucial capital and increase the risk of products becoming unsellable or requiring discount clearances, which erode profit margins. Additionally, current ordering and stocking practices may lead to lost sales due to stockouts or unnecessary costs from overstocking.

To address these issues, the project will focus on a detailed examination of sales data to better predict demand, streamline ordering processes, and minimize both stockouts and excess inventory. These efforts will help free up capital, improve cash flow, and enhance overall profitability. Furthermore, by optimizing product placement strategies based on purchasing patterns and adjusting prices in response to demand elasticity, the store can further increase sales and achieve stronger financial outcomes.

PROOF OF ORIGINALITY :

Video link:

https://drive.google.com/file/d/1N3EwAR3FWi3ZzeQVRK46a1Jm5GTBfIy-/view?usp=drive_link



Subject: Support for BDM Project - Data Sharing

Dear Ms Pammi Tiwari,

I hope this message finds you well. My name is Shivam Bhartiya, and I am the owner of Laxmi General Store. I am writing to express our full support for your BDM project at IIT Madras.

We are pleased to assist with your research by providing relevant data from our store, including details on sales trends, inventory levels, and customer demographics. Please feel free to reach out if you need any specific information or further assistance.

We are truly excited about supporting academic endeavors, and we wish you great success with your project. Do not hesitate to contact us at your convenience.

Sincerely,

Signature Shivam

Shivam Bhartiya
Owner, Laxmi General Store
Date: 03/08/2024

Figure 1: Letter from the Organization with Signature

Fig 2: With the Owner of Laxmi General Store

Fig 3 : Shop Picture

3. Meta Data and Descriptive Statistics

Dataset Title: Optimizing Retail Efficiency:A Data-Driven Approach for Laxmi General Store

Dataset Overview:
This dataset captures detailed transactional records from Laxmi General Store, offering comprehensive insights into the sales activity over a specified period. The dataset contains information related to various aspects of sales, such as the quantities of items sold, total revenue generated .

Data Source:
The dataset originates from the sales records register maintained by the owner of Laxmi General Store.

Dataset Structure and Format:

The dataset is organized in a structured tabular format with the following key columns:

1. **Date of Transaction**: The specific date on which each sale occurred.
2. **Item Name**: The name or description of the product sold.
3. **Quantity Sold**: The number of units of the product sold during each transaction.
4. **Unit Price**: The price per unit of the product sold.
5. **Sales Value**: The total monetary value of the sales, calculated as the product of the quantity sold and the unit price.

Time Frame Covered:
The data spans a three-month period, capturing transactions that occurred between March 1, 2024, and May 31, 2024.

Geographical Context:
The dataset is linked to sales activity at Laxmi General Store, which is located in the Preetam Nagar area of Prayagraj.

Column Descriptions:

1. **Date of Transaction**: This column records the exact date when each transaction was processed.
2. **Product Name**: Provides the name of the product involved in the transaction.
3. **Quantity Sold**: Indicates the number of units sold for each product within the transaction.
4. **Unit Price**: The price per unit of the product involved in the transaction.
5. **Sales Value**: The revenue generated from the sale of the specified quantity of the product, calculated as the product of the quantity sold and the unit price.

Purpose and Objectives:
This dataset is invaluable for conducting an in-depth analysis of sales patterns, customer preferences, profitability, and inventory management practices at Laxmi General Store. It serves as a crucial tool for understanding and optimizing store operations.

Key Insights Derived from the Data:

1. Identifying High-Demand Products and Peak Sales Periods: The dataset can help in recognizing which items are most frequently purchased and when the highest sales volumes occur.
2. Customer Behavior Analysis: By examining customer purchasing patterns, valuable insights into consumer preferences and habits can be gleaned.
3. Evaluating Inventory Management Efficiency: The dataset allows for the assessment of how well inventory levels are managed and their impact on sales performance.
4. Operational Performance Review: Analyzing the sales values and quantities sold provides a basis for evaluating the effectiveness of the store's operations.

Data Quality and Integrity:
Maintaining consistency and accuracy is paramount.The dataset is carefully curated to ensure uniform formatting of product names, as well as precise recording of sales values and quantities sold.

Practical Applications:
This dataset is a powerful resource for store managers and administrators, enabling them to make informed decisions regarding inventory management, pricing strategies, product placement, and overall operational improvements. The insights derived from this data can drive strategic initiatives that enhance the store's profitability and customer satisfaction.

Descriptive Statistics

The following statistics represent the analysis of transactional data from Laxmi General Store for the months of March, April, and May 2024. The data from these three months was concatenated into a single dataset to provide an overall view of the store's performance during this period.

1. Quantity Sold:

- Mean: 1.77 units
- Median (50th percentile): 1.00 units
- Standard Deviation: 4.02 units
- Minimum: 0.00 units
- 25th Percentile: 1.00 units
- 75th Percentile: 1.00 units
- Maximum: 235.00 units

2. Unit Price:

- Mean: ₹118.48
- Median (50th percentile): ₹68.00
- Standard Deviation: ₹153.85
- Minimum: ₹0.00
- 25th Percentile: ₹35.00
- 75th Percentile: ₹143.00
- Maximum: ₹5,600.00

3. Total Revenue (Amount):

- Mean: ₹150.79
- Median (50th percentile): ₹85.00
- Standard Deviation: ₹252.09
- Minimum: ₹0.00
- 25th Percentile: ₹40.00
- 75th Percentile: ₹169.00
- Maximum: ₹10,750.00

Detailed Explanation of Analysis Process/Method

The analysis of the sales data from Laxmi General Store was carried out through a systematic approach, focusing on the accurate extraction, cleaning, integration, and examination of the data. Below is a detailed explanation of the steps involved in the analysis process.

1. Pre-Analysis with Excel:

- **Data Inspection:** Before performing detailed analysis in Python, Excel was utilized for an initial review of the data. The CSV files were loaded into Excel to quickly inspect the structure, detect any immediate anomalies, and ensure the data was ready for further processing.
- **Manual Data Cleaning:** Some early-stage data cleaning was conducted in Excel, such as correcting obvious data entry errors or filling in missing values where straightforward. Excel's user-friendly interface made it easier to manually inspect and clean small portions of the data.
- **Data Mining:** Excel's filtering, sorting, and basic pivot table functionalities were employed for initial data mining. This step provided a quick overview of potential trends and patterns, which guided the more advanced analysis conducted later in Python.

2.Data Collection and Integration:

- Source of Data: The sales data was collected from Laxmi General Store's transaction records for the months of March, April, and May 2024. Each month's data was stored in separate CSV files.
- Data Loading: The data files were loaded into the analysis environment using Python's pandas library. The datasets included the following columns: Date, Item Name, Quantity, UnitPrice, and Amount.
- Data Concatenation: After loading the datasets for each month, the data was concatenated into a single DataFrame to allow for a unified analysis. This process involved stacking the datasets vertically, ensuring that the records from March, April, and May were combined while preserving the structure and integrity of the data.

3.Data Cleaning:

- Handling Missing Values: The first step in data cleaning involves checking for and addressing any missing values. Missing values in crucial columns like Quantity, UnitPrice, or Amount were identified and handled by either filling them with appropriate defaults or by removing records with missing values, depending on the context.

- Data Type Correction: Ensured that each column had the correct data type, e.g., Date was in datetime format, Quantity and UnitPrice were numeric, and Amount was calculated as Quantity multiplied by UnitPrice where necessary.
- Duplicate Removal: The dataset was scanned for any duplicate entries, which were then removed to ensure the accuracy of the analysis.

4.Descriptive Statistics Calculation:

- Summary Statistics: For a comprehensive understanding of the data, summary statistics such as mean, median, standard deviation, minimum, and maximum values were calculated for key columns like Quantity, UnitPrice, and Amount.
- Percentile Calculations: The 25th, 50th (median), and 75th percentiles were calculated to understand the distribution of data within the dataset, providing insights into the central tendency and spread.

5.Sales Analysis:

- Grouping Data by Item Name: The sales data was grouped by the Item Name, and the total revenue for each item was calculated.
- Revenue Contribution: The total revenue contribution of each product was computed to identify the most profitable items. Products like beverages, groceries, and others were analyzed in depth to understand their impact on overall sales.

6.Data Visualization:

- Histogram of Sales Distribution: Histograms were created to visualize the distribution of Quantity Sold, UnitPrice, and Amount across the entire dataset. This helped in identifying the common price ranges and sales volumes.
- Bar Charts: Bar charts were utilized to compare the total sales by category, providing a visual representation of which categories dominated the revenue.
- Pie Charts: Pie charts were used to show the proportion of sales contributed by each category, highlighting the distribution of revenue across different product segments.

7.Interpretation of Results:

- **Revenue and Sales Trends**: The analysis provided insights into sales trends over the three months, identifying high-performing products and periods of increased sales activity.
- **Operational Efficiency**: By analyzing the variance in Quantity Sold and Unit Price, we were able to identify inconsistencies and suggest potential areas for operational improvements.

- **Strategic Recommendations**: Based on the analysis, strategic recommendations were made to optimize inventory management, pricing strategies, and promotional activities.

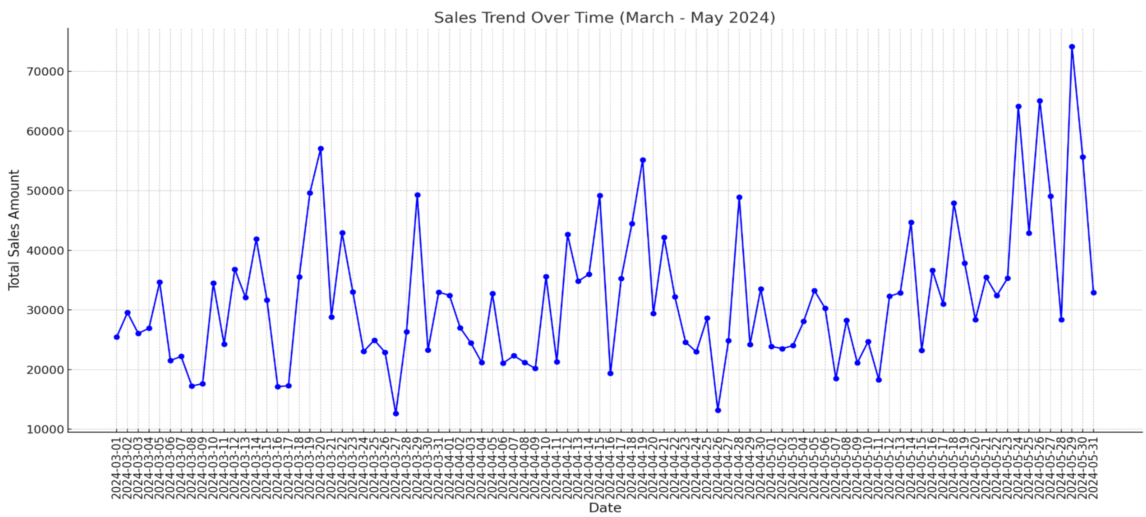
8.Tools and Libraries Used:

- **Excel:** Used for preliminary data inspection, manual data cleaning, and initial data mining.
- **Python:** The primary programming language used for the analysis.
- **Pandas:** For data manipulation, including loading, cleaning, and summarizing the data.
- **Matplotlib and Seaborn:** For creating visualizations that aid in the interpretation of the data.

The detailed analysis process involved multiple stages, from data collection and cleaning to statistical analysis and visualization. This comprehensive approach ensured that the findings were accurate, actionable, and aligned with the operational goals of Laxmi General Store. The insights derived from this analysis can be used to inform decision-making processes, optimize sales strategies, and improve overall store performance.

5. Results and Findings (Graphs and other Pictorial Representation and with words)

Graph 1:Sales Trend Over Time (March - May 2024)



This line graph illustrates the total sales revenue over time, tracked daily from March to May 2024.

- **X-Axis (Date)**: Displays the timeline from March 1st to May 31st, 2024.
- **Y-Axis (Total Sales ₹)**: Represents the total sales revenue in Indian Rupees (₹).

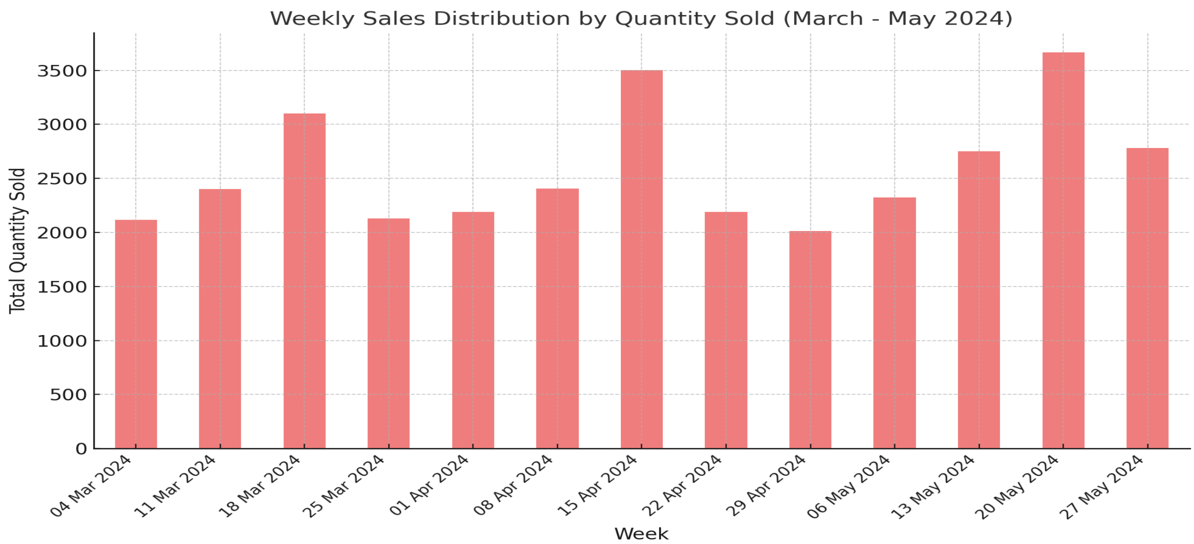
- **Blue Line with Markers**: Indicates how the sales have fluctuated over time. Each marker is a specific day, and the connected lines show the trend.

Sales Trends:

1. **March**: Sales were steady with minor fluctuations, indicating consistent purchasing behavior among consumers.
2. **April**: The shopkeeper mentioned offering discounts in mid-April to clear accumulated stock and products nearing their expiration date. This led to a noticeable spike in sales, pushing them slightly higher than in March.
3. **May**: A sharp increase in sales is observed towards the end of May as the shopkeeper again introduced heavy discounts to clear old accumulated stock and move products close to expiry.

This chart effectively shows the dynamic nature of sales over time, allowing for the identification of key periods where sales peaked or declined, which can inform future marketing and inventory strategies.

Graph 2: <u>Weekly Sales Distribution by Quantity Sold (March - May 2024)</u>



X-Axis: Weeks (March - May 2024)

- Each bar corresponds to a specific week, with dates listed along the x-axis.
- It covers the period from **4th March 2024** to **27th May 2024**, showing the total number of items sold in each week.

Y-Axis: Total Quantity Sold

- The y-axis shows the total number of units sold for each week.

- The heights of the bars represent the volume of sales in terms of the number of items.

Sales Trends:

- **March 2024**: Sales were relatively steady throughout the month, ranging between 2000 and 3000 units per week. The third week of March (around March 18) saw a slight increase in sales but was still within this range.
- **April 2024**: There was a significant spike in the third week of April (week ending April 15), where sales peaked above 3500 units. This week stands out as the highest-selling week in the entire period. Afterward, sales dipped slightly but remained high, averaging between 2500 and 3000 units.
- **May 2024**: Sales started at a lower point compared to the peak in April but saw a gradual increase. By mid-May, sales rose again to approach the 3500-unit mark, maintaining that level until the end of the month.The increase in sales during both April and May is due to the shopkeeper introducing discounts to clear accumulated stock.

Key Findings:

- High-demand products and peak sales periods were identified.
- Inefficiencies in inventory management were highlighted, impacting profitability.
- Discounts offered in April and May to clear out accumulated stock for products nearing expiry date led to spikes in sales.